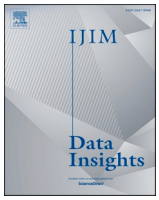
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Evaluating brand equity in the hospitality industry: Insights from customer journeys and text mining

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ABSTRACT

To achieve effective marketing, continual enhancement of product and service quality is pivotal. Assessing brand equity encompasses diverse methods, including questionnaires and interviews, unearthing invaluable customer insights. The choice of data collection methods, driven by researchers' objectives, may extend beyond direct customer interactions to employ techniques such as text mining, extracting insights from customer posts and reviews. A study at Phuket's Amari Hotel seamlessly integrated customer journey analysis with text mining, yielding a robust hotel equity model. Notably, the findings unveiled the substantial influence of factors like location and community value on hotel equity, often overlooked in previous studies. Remarkably, the hotel accommodation room emerged as the dominant touchpoint (81 %). These insights can fuel the transformation of marketing strategies and brand equity assessment. Moreover, this framework accommodates dynamic customer behavior while delivering cost efficiencies compared to alternative machine learning methods.

1. Introduction

In competitive market, marketers must align products/services with customer needs by harnessing insights from customer feedback and reviews (Li et al., 2001). However, the massive data, often in native languages on social media (Lei & Law, 2015), hindering effective brand management and profitability (Yuan et al., 2016). Business success relies on managing the marketing strategy to create customer-oriented products/services and enhance overall business performance (Sattler et al., 2010; Yuan et al., 2016). A customer-centric brand owner must focus on creating offerings with distinctive symbolic, functional, and emotional attributes (Esch et al., 2006; Keller et al., 2011; Qian et al., 2020). Effective marketing management leads to brand equity, the ultimate goal.

The management of customer review, as previously mentioned, presents significant challenges in terms of both size and complexity. Prior research has predominantly delved into the utilization of such data within linguistics or psychology, encompassing areas such as Sentiment Analysis, Text Summarization, Fake News Detection, and Fraud Detection, with a particular focus on financial contexts (Shan, 2017; Yadav & Vishwakarma, 2020; Kumar et al., 2021). Additionally, within the marketing domain, the application of this data primarily revolves around assessing the impact of product recommendations using quantitative metrics like the number of Likes, Shares, and Comments (Füller et al., 2012).

Ongoing research on brand equity is driven by technology's influence on consumer behavior. The primary framework in this field is Keller and Aker's brand equity model, which, while differing in approach, underscores the importance of brand equity in shaping consumer perceptions and preferences (Xu & Chan, 2010). Brand equity represents a brand's monetary value based on customer perceptions. For instance, well-known brands like Coca-Cola and Apple possess substantial brand equity due to loyal customers who prioritize their products regardless of pricing (2 Top Brand Awareness Models Explained, 2022).

Customer-based brand equity is a central focus in marketing, representing a product's position in the minds of consumers and creating value for both the brand and its parent company (Gordon et al., 1993; Kotler & Keller, 2005; Šerić, 2017). While extensive research has concentrated on product brand equity, service brand equity, particularly in sectors like hotels, presents unique characteristics. In the hotel industry, service consumption significantly impacts customer value, with the quality of staff service, facility cleanliness, and available amenities playing crucial roles (Kayaman & Arasli, 2007; Liu et al., 2017). As service providers actively engage customers in value creation, they must anticipate and fulfill individual needs to deliver exceptional

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experiences. Consequently, fostering strong customer relationships becomes imperative in the service sector, leading to loyalty and positive recommendations (Kim et al., 2008b; Huang & Cai, 2015). Many of these studies apply market and economic principles to design questionnaires, as demonstrated by Huang and Cai (2015) and El-Adly and Abu ELSamen (2018).

Mukhopadhyay et al. (2024) explored the integration of Big Data into the marketing strategies of Indian FMCG companies using the RBV framework. This integration enhances performance, competitiveness, decision-making, customer loyalty, and internal capability assessment, making a significant contribution to FMCG Equity. The primary data collection method employed was interviews. In the field of service businesses, research on "brand equity" can be broadly categorized into two groups. The first group focuses on enhancing factors within the brand equity model, including works by Kim et al. (2008b), Washburn and Plank (2002), Pappu et al. (2005), among others. The second group involves constructing causal models based on brand equity, as demonstrated in studies conducted by Liu et al. (2017) and Berry et al. (2000).

Previous studies like this research include the work by Kar et al. (2021), who employed an exploratory data analysis technique to analyze user-generated content (UGC) on the Twitter platform. Their aim was to identify potential constructs and formulate hypotheses for improving the customer service experience of tourists at various destinations. The factors used to develop the hypotheses were primarily identified through text mining of a large volume of Twitter data. Furthermore, Kar et al. (2023) conducted netnographic exploration by analyzing selected tweets and related literature to establish propositions that emphasize the influence of specific elements. Additionally, leveraging text-based synthesis and analysis in posts provides more accurate and detailed insights into the customer journey, including pain points and satisfaction touchpoints (Zomerdijk & Voss, 2010; Grewal & Roggeveen, 2020). In contrast, relying solely on questionnaires to explore customer value and measure brand equity may not accurately reflect the actual customer journey, potentially resulting in misleading marketing (Craig & Douglas, 2001; Piriyakul et al., 2022). These questionnaires may yield theoretical insights that differ from real-world experiences, leading to inaccurate data. To align marketing strategies more effectively with customer needs and experiences, it's essential to explore customer feedback (i.e., posts or reviews) that offer a comprehensive understanding of the customer journey (Halvorsrud et al., 2016).

Recognizing the importance of brand equity in marketing, this research addresses two key questions:

(1) How can we conduct customer reviews to extract the four components of the brand equity model?
(2) What is the relationship structure of the components in the brand equity model from customer reviews?

To comprehensively address the research problem, we employed feature selection to examine aspects of "Touch Point Events" in the customer journey and introduced a novel approach focusing on extracting causal links within Keller's brand equity model levels using trigger-based cues from reviews. This study utilized touchpoints extracted from the customer journey as a simple and cost-effective approach for the feature extraction process. It also employed Naïve Bayes to identify causal links, which is straightforward and easily comprehensible. Furthermore, the work process is transparent, functioning as a white-box model, which is easy to understand and can be proven (Loyola-Gonzalez, 2019). The amalgamation of these principles within this research has the potential to assist marketers in the strategic planning of information systems that are robust in any domain.

The study contributes to four keyways:

- Adapting the brand equity model for marketing information systems.
- Utilizing the new customer journey as an approach for feature extraction.
- Integrating Naïve Bayes for straightforward identification of causal links.
- Implementing a white-box model to ensure transparency and easy comprehension.

2. Literature reviews

2.1. Brand equity

Brand equity, a crucial aspect of marketing, is the perceived value of a brand in the eyes of consumers. It encompasses elements such as recognition and loyalty (Kotler & Keller, 2005; Kuhn et al., 2008). This perception significantly shapes consumer choices, guides marketing strategies, and influences collaborations. Moreover, it relates to the brand's ability to affect consumer perception, memorability, and recognition (Hutton, 1997). Previous research underlines the role of positive consumer experiences in nurturing brand equity by fostering loyalty, bolstering reputation, and enhancing credibility (Kuhn et al., 2008).

In the realm of brand equity, the Aker and Keller models have provided foundational frameworks for studying brand value and developing brand equity models (Keller et al., 2011; El-Adly & Samen, 2018). These models are widely acknowledged for their effectiveness in establishing and fortifying brand equity.

Keller's Customer-Based Brand Equity (CBBE) Model, conceptualized by Kevin Lane Keller, a prominent marketing professor, is a prevalent tool for brand management (Keller et al., 2011). This model employs a structured pyramid with four tiers and six essential building blocks that form the basis of a prosperous brand. These building blocks encompass brand identity, focusing on generating awareness and recognition; brand meaning, which involves establishing attribute and benefit associations; brand responses, with a focus on cultivating favorable customer attitudes and judgments; and brand relationships, emphasizing the development of profound connections that facilitate brand resonance (Huang & Cai, 2015). Proficiency in comprehending and addressing these levels and building blocks empowers organizations to construct robust brand equity.

The following sections will provide a detailed analysis of each model step and practical strategies for enhancing brand equity.

The Brand Equity Model closely aligns with the information management paradigm. The first step involves establishing brand identity and awareness, creating "brand salience" (Keller, 2008). Effective information management is crucial, leveraging market research and data analytics to understand the target audience and their needs (Prasad & Dev, 2000). Employing information management, companies identify how customers make purchasing decisions, ensuring their brand stands out and communicates a clear unique selling proposition (USP) (Pappu et al., 2005; Kim et al., 2008b). Ongoing data analysis ensures customers perceive the brand as desired, allowing for adjustments in product, service, or brand messaging.

In the second step, organizations communicate the brand's meaning, focusing on "performance" and "imagery" dimensions (Jani & Han, 2014). Information management is crucial here too. Performance involves how well the product meets customer needs, including primary characteristics, reliability, service effectiveness, design, and price (Sweeney & Soutar, 2001; Pappu and Quester, 2008). Robust information management strategies are needed to measure and enhance these attributes, involving continuous data gathering and analysis for assessing product quality, customer satisfaction, and competitive positioning. Imagery relates to how the brand satisfies social and psychological needs, influenced by targeted marketing and word of mouth (Sheth et al., 1991; Davis & Hodges, 2012). Information management tracks customer experiences, ensuring alignment with product performance, ultimately building, and maintaining brand loyalty (Hwang & Han,

2014; Jin et al., 2016).

In the third step, organizations delve into understanding how customers think and feel about their brand, closely aligning with deep analysis in information management (Kim & Kim, 2005; Kim et al., 2008b). This step considers two categories of customer responses: judgments and feelings. Judgments are rooted in quality, credibility, consideration, and superiority, and organizations employ information management techniques to improve these aspects. Strategies include enhancing product quality, building credibility through expertise and trustworthiness, addressing customer needs, and evaluating competitive positioning (Netemeyer et al., 2004; Kim & Kim, 2005). Additionally, understanding customer emotions and the six positive "brand feelings" (warmth, fun, excitement, security, social approval, and self-respect) is essential for information management. The concept of prestige value, a crucial factor in brand resonance (Hwang & Han, 2014), requires a nuanced approach to information management that resonates with customers.

The final step is creating powerful emotional connection, known as "brand resonance." Marketers strengthen this objective by harnessing technology for customer relationship management through social media (Salloum et al., 2017a). Brand resonance comprises four elements: behavioral loyalty, attitudinal attachment, a sense of community, and active engagement. These elements benefit from effective data-driven strategies like loyalty programs, social media engagement, referrals, and customer feedback analysis (Kar et al., 2023). Data management collects insights into customer behavior, enabling resonant strategies (Keller, 2003; Huang & Cai, 2015).

Aaker's brand equity model, a well-recognized framework in brand management, complements Keller's model. Aaker's approach focuses on brand equity as a combination of assets and liabilities associated with a brand, influencing its overall value (Aaker, 1991). This model deconstructs brand equity into five key components: brand loyalty, brand awareness, perceived quality, brand associations, and other proprietary brand assets. Emphasizing the creation of a distinctive and favorable brand identity, Aaker underscores the cultivation of robust brand associations that differentiate one brand from competitors (Aaker, 1991; Bendixen et al., 2004). Additionally, an efficient information management system aligns with proactive brand equity cultivation, enabling companies to enhance competitiveness, build strong customer loyalty and advocacy, and achieve long-term growth (Sweeney & Soutar, 2001).

This research is reminiscent of the study conducted by Kushwaha et al. (2021), which aimed to propose and test a theoretical model designed to address the identified gap related to customer experience as a differentiating factor between interactions involving customer agents and those involving chatbots.

The brand equity model, originally from marketing, has significant potential for contributing to Information Systems (IS) literature. Its adaptation to IS contexts can provide insights into user perceptions and interactions, influencing adoption and loyalty. The model's dimensions, including brand awareness, associations, quality, and loyalty, offer valuable insights into determinants of IS adoption. Exploring brand resonance enhances the IS user experience and engagement. Applying this model reveals causal relationships among system components, identifying critical factors for IS success.

2.2. Hotel industry and hotel equity

The hotel industry is a significant part of the service industry, providing temporary accommodation and a range of amenities and services to paying guests. As of 2019, it had reached a peak market size of 1.52 trillion US dollars, making a major contribution to employment in the business sector. However, due to the COVID-19 pandemic, the market size dropped below one trillion dollars in 2020 and 2021, and it is projected to recover to 1.06 trillion dollars in 2022 (Statista, 2023). Despite this setback, it is expected that the hotel industry in various countries will recover and generate significant revenue for their respective GDPs in the post-pandemic era. The Thai Ministry of Tourism reports that the tourism industry will bring in between 1 and 1.5 trillion baht in 2022 (Tourism Statistics Thailand, 2000–2022).

In the realm of hotel accommodations, a diverse array of amenities and services caters to paying guests. Hotel proprietors often aim to attract specific customer segments through pricing, marketing strategies, and service selection. According to a review outlining the essential features of an exceptional business hotel (Tucke, 2015; Kumar et al., 2021), a hotel should offer nine key elements to its clientele. These include Wi-Fi connectivity, well-equipped workspaces, fitness facilities, communal areas, express check-in/check-out options, and a prime location. Fundamental hotel services typically comprise guest reception, room service, dining facilities and security measures. Numerous studies have explored customer value within the hotel industry, such as those conducted by Heung and Lam (2003) and Kandampully and Suhartanto (2000). In the telecommunications industry, this study highlights the transformative impact of big data analytics on decision-making for product-service bundling strategies, ultimately enhancing Brand Equity (Zakaria, 2024; Kimpakorn & Tocquer, 2010).

In the context of the hotel equity, previous research was conducted on a variety of study theme lists in Table 1.

Based on the research findings presented, here are three hypotheses that can be formulated:

H1-H3: The use of hotel amenities by a review writer significantly affects the writer's perceptions of hotel image, emotion, and resonance.

2.3. Customer journey

A customer journey (CJ) is the sequence of interactions a customer has with your company (Følstad et al., 2013; Dove et al., 2016). Touchpoints can vary from the time they search for you on Google to the time they use one of your products/services. A touchpoint is any time a potential customer or customer encounters a brand—before, during, or after they purchase something from the brand owner (Følstad et al., 2013). When it comes to traveling and staying in a hotel, every interaction with the customer, whether within or outside the hotel, is a

Table 1
Previous Studies on Brand Equity.

Authors	Research results
Berry (2000)	In service organizations, the research suggests that the main sources of brand equity are "brand meaning," shaped by customer perceptions and experiences.
Xu and Chan (2010)	Research findings show that Customer-Based Brand Equity (CBBE) in hotels arises from customers' direct experiences through service performance and their indirect experiences through advertising and word-of-mouth.
San Martín and Del Bosque (2008)	The research results indicate that cognitive affective psychology plays a vital role in the emotional response of hotel guests.
Kim et al. (2008b)	The research investigates the impact of brand equity on guests' perceived value, cost considerations, and intention to revisit mid-priced hotels.
Liu et al. (2017); Hsu et al. (2012)	The research shows that three dimensions of Customer-Based Brand Equity (CBBE) - brand loyalty, perceived quality, and brand image - significantly affect purchase intention in luxury hotels. Brand awareness, on the other hand, has no significant impact.
Shan (2017)	The research developed a luxury and upscale hotel sector-specific CBBE model. It identifies key factors for brand equity in this sector: brand image, customer relationship management, social image congruence, brand affect, brand trust, and consumer-generated content.
El-Adly and ELSamen (2018)	The GBHE research enhances hotel equity measurement by incorporating cognitive and affective dimensions to capture the complete guest experience, including factors like price, quality, self-gratification, aesthetics, prestige, transaction, and hedonism, contributing to customer perceived value.

touchpoint. Each touchpoint involves a service provided by the hotel and an action taken by the customer. If there are any issues or obstacles, also known as pain points, they can prevent customers from taking necessary actions (Crosier & Handford, 2012). One common pain point is a defect or fault in the service. Therefore, CJ, as essential tools in service management, describe how guests engage with a service brand from initial awareness through booking and their stay. These journeys are visualized through customer journey maps (CJMs), depicting various stages of interaction, including touchpoints like website interactions, or booking with hotel representatives (Baye et al., 2016). Modeling guest behavior within these journeys is crucial for enhancing guest satisfaction and revenue. CJMs typically include touchpoints, stages, and emotional aspects, like satisfaction or dissatisfaction. There is no standard approach to constructing CJMs, with hotels using quantitative or qualitative analysis (Bernard & Andritsos, 2017). Research on CJMs covers topics such as service design, guest experience integration, and co-design. Guests' emotional, social, and spiritual responses occur at specific touchpoints, either directly through physical interactions or indirectly through their perception of the hotel's value proposition via various channels. Improving the guest journey involves reducing pain points and considering cross-cultural differences to enhance brand equity (Baye et al., 2016).

Regarding hotel reviews, CJ includes both purchase touchpoints and post-purchase touchpoints. During the service, customers report their experiences. It is essential to optimize purchase touchpoints to ensure that customers have a positive, consistent, and relevant experience, which increases the likelihood of repeat business and referrals. Post-purchase touchpoints, on the other hand, refer to customer interactions that occur after a purchase, such as a return or a recommendation.

The study's conclusion presents an event at each touchpoint using the following mathematical model:

$$TPEi = (TUi, SAi, PIi, JFi, BRi); i = 1, 2, \ldots \quad (1)$$

Eq. (1) expresses the touchpoint event as TPEi, which can occur either inside or outside the customer's environment. TUi denotes the type of touchpoint, while SAi, PIi, JFi, and BRi represent features that align with the hierarchical level of brand equity (i.e., SAi represents the Salient feature from touchpoint type "i"). This step uses touchpoints related to activities within the hotel and outside. Use it as a bait to find the location of words in a document. Then select a group of consecutive 6 words. This step involves extracting relevant features, namely SA, PI, JF and BR.

Tourist hotel guests seek more than comfort and amenities; they desire local experiences. Hotel value extends beyond facilities, considering location and surroundings. Crafting a CJ via reviews enhances brand resonance, leading to a new framework for hotel equity exploration.

Human communication patterns influence travel recommendations, often linked to psychological traits (Fischer et al., 1990) and supported by the appraisal theory. When individuals encounter real-world experiences, they seek meaning, make inferences, and appraise them, triggering emotions that shape their actions. Jha and Shah (2019) suggest that the decision to write a review and its content are influenced by a multi-step, nonlinear appraisal mechanism. The initial step involves self-evaluating one's own experience, generating positive or negative sentiments that influence the decision-making process. Research indicates that individuals share up to 90 percent of their emotional experiences (Rimé, 2009), providing a vital channel for expressing feelings, their reasons, and intended actions. Sharing emotions helps individuals manage their well-being by addressing external stimuli-induced emotions (Berger, 2014).

This paper adopts the social appraisal theory as a lens to gain insight into the cognitive and psycho-social mechanisms that influence the decision-making process of review writers, including what and how they write. To achieve this, we draw from literature in marketing, information systems, and organizational behavior. The theory provides a suitable framework for understanding the behavioral aspects of review

H4-H5: The hotel image portrayed by a review writer significantly affects the writer's perceptions of emotion and resonance.

H6: The emotional tone conveyed by the reviewer significantly influences the writer's perceptions of resonance.

2.4. Knowledge mining from textual data

With the explosive growth of user-generated texts on the Internet, the automatic extraction of useful information from abundant documents receives interest from researchers in many fields, in particular the community of Natural Language Processing. Opinion mining (Radice, 2018) was first proposed in the early years of this century and has since gained popularity as an active research area. Moreover, various practical applications of opinion mining, such as product pricing, competitive intelligence (Xu et al., 2011), and risk detection in banking systems, draw extensive attention from industrial communities. Knowledge extraction from language, also known as text mining, is a widely used approach in numerous fields, including health care, government, education, and manufacturing. The text mining process utilizes technologies such as pattern matching, topic tracking, summarization, clustering, and information visualization (Salloum et al., 2017b). Pattern matching involves analyzing unstructured text and identifying key phrases and relationships within the text by looking for predefined sequences.

This study aims to employ text mining techniques for extracting semantic characteristics from review texts in the domain of tourist accommodation and travel experiences. As demonstrated in Cao et al.'s work (2011), these semantic characteristics significantly impact the number of helpfulness votes received by reviews compared to other features. Therefore, correct attribution of features and classifications to text is crucial, as incorrect attribution can yield meaningless results. The widely used technique for feature selection is the "bag of words" method (Da Silva et al., 2014), which involves breaking down text into words, treating each word as a feature, and counting its frequency of occurrence while disregarding co-occurrence (Wartena et al., 2010). Additionally, it is imperative to investigate and identify significant causal words.

2.5. Sentiment analysis

Sentiment analysis (SA) is a technique used to categorize and evaluate opinions expressed in large amounts of online data (Lee, 2016; Ameen et al., 2021; Amin-Nejad et al., 2020). It examines the author's attitude towards a particular topic, such as a movie, product, or service. Online communication sites and social media platforms generate vast amounts of data containing people's experiences, feelings, and thoughts, which can provide valuable insights for decision-making, particularly in service industries and online marketing (Kumar et al., 2021). Automated content analysis, utilizing natural language processing, text processing, and machine learning, is often employed to extract information from this data (Mohey, 2016). The authors applied SA at two levels: firstly, at the whole review level to cluster customer segments, and secondly, at the phase analysis level to measure the significance of touchpoint events.

2.6. Causality extraction

Causality, in the context of natural language, refers to the relationship between cause and effect as expressed through language and communication. It involves understanding how one event, action, or condition leads to another, often described through linguistic cues and patterns. In natural language, causality can be inferred and expressed in various ways: (1) Causal Connectives: Words and phrases like "because," "since," "due to," and "therefore" are used to indicate causal relationships between events or ideas. (2) Conditional Statements: Conditional language, such as "if…then," is commonly used to establish causal links. For

instance, "If it rains, then the ground will be wet." (3) Verb Tense and Modifiers: Verb tenses like past perfect ("had caused"), past simple ("caused"), and present perfect ("has led to") can imply causality. Additionally, adverbs like "directly," "indirectly," "consequently," and "subsequently" help specify causal connections. (4) Sequential Order: The order of events in a narrative or description can suggest causality. When one event follows another, it may imply a cause-and-effect relationship. (5) Co-Occurrence and Correlation: Observing two events or conditions occurring together or in a consistent pattern may imply a causal link. However, correlation does not always indicate causation, and additional evidence is often required to establish causality. And (6) Rhetorical Devices: In persuasive communication, authors may use rhetorical devices like persuasive appeals (ethos, pathos, logos) to persuade the audience of a causal relationship between their arguments and desired outcomes (Radinsky et al., 2012; Xu et al., 2020).

Establishing causality from natural language often involves careful analysis, critical thinking, and consideration of context. It's essential to recognize that correlation, temporal proximity, and linguistic cues can suggest causality, but rigorous scientific methods are needed to confirm causal relationships in many cases. This study integrates method 1- 5 with consistent to the message as simple narrative language to explore the causality and further testing by using Bayes judgment.

Causality represents a kind of relationships between cause and effect, stating that the happening of causes will trigger the happening of effects, which constitutes the basics for inference and reasoning. In early days, people found causal relations from production process and daily lives manually, slow, and inaccurate. The ever-growing web resources have made it feasible to automatically extracts causality from text, triggering an emerging and hot topic in NLP, with abundant downstream application tasks such as event detection and prediction, questions answering (Hashimoto et al., 2014; Radinsky et al., 2012).

3. Data collection

The study collected hotel review data from TripAdvisor to identify the most highly reviewed hotels in Phuket in 2022. The tourism industry has been a significant contributor to the economy of Phuket, as evidenced by the fact that it generated 233.38 billion baht in tourism revenue in 2022. While this figure is lower than before the Covid-19 pandemic in 2019, where the annual tourism revenue stood at 400 billion baht, foreigners still make up more than 70 % of the total revenue, while the remaining is from Thai visitors. This highlights the importance of hotel selection in Phuket to the tourism industry. The average hotel room occupancy rate was 42.58 % last year, but currently, the rate is 80 % out of 101,221 rooms available. In terms of foreign arrivals, Russia was Phuket's biggest tourist group, followed by India, Australia, the United Kingdom, Singapore, Germany, Malaysia, Kazakhstan, South Korea, and the United States. Understanding the preferences and needs of these foreign visitors in hotel selection is crucial for maintaining and increasing the tourism revenue in Phuket (Phuket tourism revenue moving closer to pre-pandemic level, 2023).

During the year 2022, the Amari Phuket hotel was ranked by 463 hotel customers in the 4th and 5th levels, and the hotel received a 4.5 rating from a total of 9695 reviewers (TripAdvisor.com, February 15, 2023). Since, the study aims to explore brand equity based on a theoretical foundation and previous research, we randomly selected 250 reviews from the Amari Phuket hotel (maximum reviews in TripAdvisor.com 2022) for further analysis. Likewise, this led to the use of "Amari equity" instead of "hotel equity" in the next section.

The operational procedure is categorized into four distinct parts:

Part 1: Tokenization, lowercasing, stop word removal, stemming, and lemmatization of the data from the review message.

Part 2: Sentiment analysis using dictionary base software to extract sentiment and magnitude value.

Part 3: Identifies elements of different types of touchpoint events and their associated features by utilizing customer journey in a hotel. The output of this part is illustrated through an example in the following structure.

A review is depicted as follows:

{($TU_{i,j}$, ⟨$SA_{i,j}$⟩, ⟨$PI_{i,j}$⟩, ⟨$JF_{i,j}$⟩, <$BR_{i,j}$, ($TU_{i,k}$, <$SA_{i,k}$>, ⟨$PI_{i,k}$⟩, ⟨$JF_{i,k}$⟩, <$BR_{i,k}$}; where "i" signifies the review number, while j and k represent two distinct touchpoints. It's important to acknowledge that a single review may encompass multiple touchpoints, as is the case when a customer journey includes experiences related to both the restaurant and front-end service. Following this, the system combine $SA_{i,j}$ and $SA_{i,k}$ into a unified SA_i, and apply a similar merging process to the remaining sets of data. As a result, the data structure of this review can be represented as a vector containing ($TU_{i,j}$, $TU_{i,k}$, SA_i, PI_i, JF_i, BR_i); where i represents the review, while j and k denote distinct touchpoints.

Part 4: During this phase, the system extracts nouns, noun phrases, verbs, and verb phrases from the surrounding 6-grams of data using a predefined set of cause-and-effect cue words, as discussed in the Knowledge Mining from Textual Data section. This process combines keywords and human review to extract knowledge related to cause and effect. Subsequently, iterative procedures involving Naive Bayes are employed, applying the principle of posterior probability with a 10 % threshold to identify cause-and-effect relationships based on rule-based judgments. Additionally, Naive Bayes is utilized for category analysis based on word frequency to explore the relationships between "Identity," "Meaning," "Response," and "Relation" through customer reviews.

To validate and confirm the causality model, sentiment values at the phase level (SA_i, PI_i, JF_i, and BR_i, as outlined in part 3) are incorporated and formulated to run as a structural equation model (SEM).

4. Methodology

4.1. K-Mean

One of our objectives, utilizing data from *part 3*, is to segment customers by employing sentiment and magnitude variables extracted from each review. These data will undergo analysis using the K-means technique. Before implementing the algorithm for this method, the dataset will be visualized to estimate "k," which tentative represents the number of predefined groups.

The segmentation algorithm operates as follows:

Given an initial set of k centroids (means): $C_1^1, C_2^1, .. C_k^1$ the algorithm alternates between two steps:

1. Assignment step: Each observation is assigned to the cluster with the nearest mean, determined by the least squared Euclidean distance.

$S_i^{(t)} = [x_p : \| x_p - C_i^{(t)} \|^2 \leq \forall_j, 1 \leq j \leq k]$; where each x_p is assigned to one $S_i^{(t)}$

1. Update step: the centroid of each cluster is updated after each data point is assigned to that cluster. The recalculated centroid is given by $C_i^{(t+1)} = \frac{1}{|S_i^t|} \sum_{x_j \epsilon S_i^{(t)}}^{t} x_j$

Studies leveraging big data frequently involve grouping the data into clusters or segments to derive insights across various domains, such as market segmentation. Fundamentally, this approach necessitates the utilization of quantitative data as its input (Huang et al., 2007; Bansal et al., 2017.

4.2. Naive bayes classifier

Naive Bayes classifiers, based on Bayes' Theorem, are a group of algorithms known for their simplicity and effectiveness in classification tasks. They assume the independence of features and are particularly useful for fast predictive modeling in machine learning. In this study, the

Naïve Bayes classifier was chosen to assess whether trigger words could determine the reasonableness of customer reviews. The key principle behind using the Naïve Bayes classifier is that data features are conditionally independent of each other, given the class label, and the data is qualitative, involving the number of words extracted from the document.

The Naïve Bayes algorithm has garnered significant attention from researchers, attributed to its demonstrated capability in tackling intricate real-world challenges (Da et al., 2011). While Da et al. (2011) highlighted its utility in forecasting trends and facilitating decision-making processes within distance education systems, Neogi et al. (2021) aimed to comprehend the sentiments of Indian citizens towards three specific governmental acts through the integration of natural language processing techniques. Furthermore, their research endeavored to analyze the polarity and factuality of Twitter data pertaining to the associated demonstrations by extracting relevant tweets and employing Naive Bayes classifiers in conjunction with sentiment analysis methodologies.

The classification is determined by the formula: $\hat{y} = \text{argmax}_x\ P(y) \prod_{i=1}^{n} P(x_i|y)$, where $\hat{y}$ represents the class label that maximizes the expression. P(y) typically denotes the proportion of class "y" in the training set, also referred to as the "prior," while $P(x_i|y)$ represents the probability of observing the feature value x_i if the true class is indeed y, known as the "likelihood."

4.3. Jaccard similarity

The results obtained through the Naïve Bayes classifier follow this format: <cause_1, effect_1>, …. ⟨cause_k, effect_k⟩. To identify which cause_k and effect_k correspond to SA, PI, JF, and BR, the qualitative data of cause_i and effect_i are processed using specific equipment. Jaccard similarity is then applied.

The Jaccard similarity index primarily considers global ratings. It calculates the ratio of the shared items' count to the total count of items rated in both sets.

$Sim(u,v) = \frac{(I_u \cap I_v)}{(I_u \cup I_v)}$; where I_u is "cause_i" and I_v is a set of SA, PI, JF, and BR

If the similarity value between I_v and each set of SA, PI, JF, and BR is the highest, it indicates that I_v is similar to that group. This also applies to "effect_i." For instance, in the sentence "*Good service* leads to *Amari advocacy*," "*Good service*" represents the cause (SA) and "*Amri advocacy*" corresponds to the effect (BR). When seeking similarity, we associate "*Good service*" with SA and "*Amari advocacy*" with BR. Consequently, we establish an influence line weight from SA to BR, denoted as $SA \rightarrow^1 BR$, through this process. The outcome of this work is to visualize the relationships between the components of brand equity in a graphical format for further testing hypotheses.

5. Analysis and results

Since Amri Hotels received a total of 488 reviews in 2022, a sample size of 250 reviews is considered statistically reliable. The minimum sample size required to achieve a margin of error of less than 5 % and a confidence level of 95 %, as recommended by Yamane's formula, is 220 units (Tepping, 1968).

Part 1: Data preprocessing

Data from 250 reviews will be used to cut through the noise (using non-language symbols) and processed in the next step. Due to the work at this stage, the language has been reformulated, which may cause confusion. Therefore, the performance report will still use the English language that appears in the document for comprehension compared to the original document.

Part 2: Segmentation Analysis for Amari Resonance.

The sentiment and magnitude values of 250 reviews are listed in Table 2.

Table 2
Sentiment and magnitude value of the sample data.

Review	Length (characters)	Sentiment value	Magnitude value
1	283	1.00	2.63
2	503	1.00	4.25
3	231	1.00	2.43
4	503	0.71	4.62
5	503	−0.58	2.18
6	215	0.75	1.88
7	330	0.14	3.42
8	340	0.09	3.19
9	503	0.18	3.14
10	483	0.70	3.2
11	503	0.56	3.06
12	328	0.64	1.64
13	503	0.69	1.96
:	:	:	:
:	:	:	:
250	225	0..69	2.5

The sentiment value ranges from −1 (negative) to 1 (positive), while the magnitude value starts at 0. The sentiment score of a document is an indicator of the overall emotion expressed in the document, while the sentiment magnitude indicates the amount of emotional content present in the document. This value is typically proportional to the length of the document. These two values can be leveraged as an indicator of customer loyalty as well as an indicator of which customers are less likely to return for future visits.

The K-means clustering analysis conducted with sentiment and magnitude values revealed that customers can be classified into four distinct groups based on their loyalty toward the hotel, measured by their emotional and magnitude scores (Fig. 2). The first group, accounting for approximately 38 % of customers, shows high loyalty (high resonance) and is indicated by an orange circle. The second group, representing 52 % of customers, displays moderate loyalty (middle resonance) and is marked with a black circle (low resonance). The third group, comprising 9 % of customers, has low loyalty and is represented by green circles. The fourth and final group, accounting for 1 % of customers, has expressed negative opinions and is unlikely to return to the hotel (churn). This group is marked with a red circle.

Part 3: Touch point event construction

The objects that meet the TPEi component requirements (as referred to in Eq. (1)) and are deemed touchpoints in the sample message have been examined, and the resulting lists can be found in Table 3.

After customers make a purchase, their reviews tend to focus on the significant characteristics of each touchpoint experience, which are strongly imprinted in their memory. This simplifies customers' perception of Amari's brand into four dimensions, including "brand identity," "brand meaning," "brand response," and "brand loyalty." Additionally, the review highlights the significance of the Amari Hotel's name, which is mentioned in 53 % of the reviews. The results presented in Table 4 indicate that "location value" and "hotel neighborhood value" are mentioned more than 20 % of the time, indicating the emergence of new perceived values within the scope of brand equity. Hence, a customer journey map for hotel management is depicted in Fig. 3. It classifies touchpoints (TU_3) after the check-in process with the front-end service into two categories: inside and outside the hotel. Among these touchpoints, TU_6 is the most frequently recommended, representing 81 % of the sampled data.

Part 4:

To compare with questionnaire-based tasks, we aim to identify cause-and-effect relationships and perform posterior probability instead of using a quantitative regression analysis. Since our data model consists of unstructured data, we employ a keyword selection approach to extract relevant information related to Keller's brand equity level. This step is divided into two distinct tasks for analysis.

Task 1: The use of the term "recommend" in a review after a travel

Table 3
List of touchpoint events.

TU	SA	PI	JF	BR	%
Hotel location (TU_1)	Beauty beach, Excellent location, Panaroma view, Hospitality ecosystem	Amazing to relax Happy time Great experience	Couple hotel, Family hotel, Romantic hotel	Revisit Amari advocacy	91.2
Front end service (TU_2)	Fulfill, speedy, empathy, friendly	Worth welcome	Service quality	Word of mouth	48.00
Hotel restaurant (TU_3)	Varity of food, Fresh food, Great quality	Fresh, Delicious, Quality, Good, Great experience	impressive	Revisit Word of mouth	36.50
Outside restaurant (TU_4)	Fresh sea food, Good taste, Delicious, Amazing Thai food	Fresh, Delicious, Quality, Good, Great experience	impressive	Revisit Word of mouth	57.20
Breakfast point (TU_5)	Good service, good place, excellent food, and drink	Good quality, Excellent	Enjoy Impressive	Word of mouth Amri advocacy	67.20
Room, -room (TU_6)	Clean, Luxury, Lovely, Excellent location, Hygienic room, Excellent accommodations, Fulfilled with amenities	High quality, Over expectation	Enjoy, Happy, Relaxing	Revisit Word of mouth	80.80
Spa, Massage (TU_7)	Clean, Hight service	Best quality, Responsive, Empathy	Enjoy, Happy, Relaxing, Good experience	Revisit Word of mouth	31.20
Swimming pool (TU_8)	Area was amazing, Near beach, Great	Convenience	Enjoy, Happy, Relaxing	Word of mouth	18.40
Club lounge (TU_9)	Good service	High satisfaction	Enjoy, Good relaxing	Word of mouth	46.40
Beach (TU_{10})	Excellent, Clean, Excellent view, suitable for a walk	High satisfaction	Enjoy, Happy, Relaxing	Revisit, Word of mouth	56.80
Hotel neighborhood (TU_{11})	Good, Friendly, Good welcome	Satisfaction	Good experience	Word of mouth	26.00
Airport (TU_{12})	Convenient transportation, good service, fast	Satisfaction		Word of mouth	4.80

Note: "-room" refers to the bedroom or bathroom; percentage calculated based on frequency of occurrence based on sample size.

Table 4
Examples of word sequences that come before and after the word 'recommend' in the sentence.

Review	Sample
1	I fully *recommend* this hotel location is great for
2	Would definitely *recommend* to other friends.
3	Additionally, I would *recommend* improving the wifi in the hotel rooms.
4	We highly *recommend* this hotel for anyone visiting Phuket
5	of which La Gritta is highly *recommended*
7	Totally *recommended.*
8	I would highly *recommend* trying out their spa treatment.
9	We would stay again and *recommend* to family and friends visiting Phuket
10	Not *recommended* for people with physical disabilities or
11	Highly *recommended.*
12	Best hotel ever highly *recommended* for family vacations.
13	One bedroom balcony suite - really *recommend.*

Note: If any sentences are short in length, they may not contain the requisite six words before or after the focal word.

experience at Amari Phuket is supported by self-efficacy theory and social communication theory. The term implies a level of confidence in one's ability to communicate positive experiences to others, reflecting the knowledge and experiences of consumers. According to self-efficacy theory, an individual's belief in their ability to perform a task effectively influences their behavior. In this context, the use of the word "recommend" serves to enhance the reviewer's self-efficacy as they perceive themselves as knowledgeable and effective communicators of their experiences (Orth & Robins, 2014). Moreover, social communication theory suggests that social interactions and communication play a significant role in shaping individuals' attitudes and behaviors. As consumers share their positive experiences with others, they contribute to the formation of positive attitudes towards the brand, which can influence others' purchase decisions. Thus, the use of the term "recommend" in the context of travel reviews reflects the interplay of personal experiences, individual beliefs, and social communication in shaping consumer behavior. Identify the six-gram word sequence that contains the term "recommend" and extract the list of features from its pre- or post-words, as listed in Table 4.

The extracted data was evaluated for the occurrence of six words before and after the keyword "recommend" in the same sentence. In a sentence like "Best hotel ever highly recommended for family vacations," the phrase "Best hotel ever highly" is the causal factor. If a similar relationship exists within hotel contexts, the Naive Bayes approach can be employed for further analysis. However, in some instances, such as "Additionally, I would recommend improving wifi in hotel rooms," the causal factor of "improving wifi in hotel rooms" is not true.

Analyzing the sentences in the text sample that included the term "recommend" from previous and post reviews revealed 175 statements from 250 reviews. Among these statements, 152 were considered effective and causal, whereas 23 statements, including those from reviews 3 and 4, were not related to hotel recommendations. The application of the Naïve Bayes principle facilitated the ability to make the subsequent determinations.

$$P(y=1|recommend)=\frac{152}{175}\ P(y=0|recommend)=\frac{23}{175}$$

$$\hat{y}=\text{argmax}(P(y=1|recommend),P(y=0|recommend)\ \rightarrow\hat{y}=recommend))$$

Let $y=1$ represent causality, while $y=0$ indicates the lack of causality. Therefore, within this context, if the term "recommend" appears, we can infer that a sequence of words meeting the predefined criteria implies causality. As a result, it can be concluded that the various touch points, products, and service attributes generate a positive impact on Amri loyalty. The causal elements within this scenario include the hotel location, Amri, club house, spa, room, staff, breakfast, and restaurant. Customer loyalty (word of mouth) is the result.

Task 2: The results of using keywords from the data that was

analyzed to find correlations that support the hierarchical brand equity model are shown in Table 5.

Following the argmax posterior Bayes analysis (a simple calculation) conducted in Table 5, potential causal relationships were identified among the variables listed in Tables 3 through 5. These findings are presented in Fig. 3, which provides a comprehensive view of the review population and how these relationships operate. We extracted a total of 12 touchpoints, which are divided into two categories: internal hotel touchpoints (such as hotel services) and external hotel touchpoints (such as restaurants, the beach, the hotel neighborhood, and the airport). The salience (or identity) of internal hotel touchpoints consists of aspects such as cleanliness, health, and high-quality service, while external touchpoints are characterized by features such as excellent locations and friendly environments. Additionally, our analysis revealed that the Amari brand is associated with attributes such as high quality, convenience, and high customer satisfaction, which ultimately contribute to positive customer experiences and enjoyment. Finally, the top level of the Amari equity pyramid is characterized by customer intention to revisit and positive word-of-mouth recommendations on social media.

Table 5
List of samples from the analysis of cue words and their relationship with cause and effect.

Begin cue word (effect)	Sample sentences	P (cue| cause, $y = 1$)	P (cue| cause, $y = 0$)
because	"The location is perfect in our opinion *because* it gives both the perfect sea view." ($y = 1$) "It was difficult *because* of COVID to get all paperwork required together." ($y = 0$)	0.83	0.17
due to	"*Due to* their awesome hospitality and very friendly staff, I extended my five nights stay…" ($y = 1$) "…was not able to visit last year *due to* Covid." ($y = 0$)	0.75	0.25
if	*"if you visit Phuket and stay at Amari, which offers great accommodation, you are likely to revisit and recommend it to others."* ($y = 1$) "Great spot to stay *if* you want to be in, and yet out of the hustle of Patong." ($y = 0$)	0.74	0.26
since	"Since we stay at Amari, my family is happy." ($y = 1$) "*Since* we were a family of four with two adults and two kids, we chose to book a two-bedroom suite." ($y = 0$)	0.30	0.70
contribute	*"The luxurious and warm atmosphere of Amari contributes to customers' emotional and mental well-being."* ($y = 1$) *"I believe that everyone can contribute something positive to the world,"* ($y = 0$)	0.60	0.40
lead to	"Good service *leads* to Amri advocacy." ($y = 1$) "Knee pain *leads* to where I want to go." ($y = 10$)	0.89	0.11
result	"Excellent service results in a happy vacation." ($y = 1$) "A clean room instills trust in the hotel's commitment to health." ($y = 0$)	0.75	0.25
affect	"Excellent service *affects* the happiness of a vacation." ($y = 1$) "A clean room *affects* trust in the hotel's commitment to health." ($y = 0$)	0.65	0.35
influence	"Excellent service *influences* the enjoyment of a vacation." ($y = 1$) "Crowded areas *influence* tourists' privacy." ($y = 0$)	0.53	0.47
why, reason	"… standards and security of a good hotel, that is *why* we chose to return to Amari Phuket." ($y = 1$) "Covid *19 is a reason why we did not travel.* ($y = 0$)	0.33	0.67

Remark: $y = 1$ is class true (cause-and-effect), and $y = 0$ is class fault (not cause-and-effect).

Fig. 4 illustrates that some of the reviews draw a direct line from the brand identity to the brand salience without linking to the brand meaning and response, for example, "good service leading to Amari advocacy." However, some reviews form a complete chain of brand equity, starting from the brand identity, going through the brand meaning, and resulting in customer response. The probabilities in this figure are constructed based on Table 5. If the categories are in the same meaning block, we use average scores to indicate the direction of the relationship.

To validate the graphical relationship shown in Fig. 4, the authors curated a dataset by selecting features from Table 3 (coding them to sentiment values at the phase level). This dataset was then used for subsequent analysis using structural equations. The findings, presented in Fig. 5, confirm the consistency with Fig. 4.

The model analyzed all the samples to identify causal hypotheses. The model analysis and fitting showed that the fitness index ($p = .063$, cmin/df = 1.147, CFI = 0.944, NFI = 0.953, RMR = 0.034, and RMSEA = 0.035) met the threshold for fit indices, indicating an acceptable level (Schlermelleh-Engel et al., 2003; Vandenberg, 2006).

The disparity between the model, as depicted in Fig 4, and Keller's brand equity model (Fig. 1) can be discerned. Document reviews, composed in natural language, tend to be concise, emphasizing key points while potentially omitting intermediate steps. As a result, the brand equity model derived from customer reviews of Amari hotel stays exhibits some deviations. Nevertheless, it is evident that Amari Hotel possesses brand equity. The findings of this study suggest that the management practices employed by this hotel can serve as a benchmark for enhancing the competitiveness of other establishments in the industry.

6. Discussion

6.1. Theoretical contributions

The examination of 250 customer reviews of Amari Hotels revealed the presence of four distinct customer segments, with a noteworthy 90 % displaying high loyalty and a strong inclination to recommend Amari Hotels to others. An impressive 61 % of these recommendations highlighted the hotel's strengths across various touchpoints, including special events, thereby reinforcing the hotel's positive reputation and brand equity (Keller, 2008; El -Adly & Samen, 2018; Kandampully & Suhartanto;, 2000). Additionally, these findings align with the research conducted by Prasad and Dev (2000), which elucidated the influence of various measures of brand awareness, including brand recall, top-of-mind awareness, and total brand awareness, on brand performance. Consequently, the identification of Amari Hotel as either a "Family hotel" or "Couple hotel" in this study should be regarded as a semantic or conceptual representation of the hotel's image, offering valuable insights for targeted marketing strategies (Baye et al., 2016; Bernard & Andritsos, 2017).

A significant finding in this research is the high value placed on the hotel's location, emphasizing its role as a distinctive feature of Amari Hotels. This focus on location value goes beyond previous hotel equity studies, such as Jani and Han (2014) and Kayaman and Arsali (2007), and the use of cognitive-affective psychology to assess customer-perceived hotel value, as seen in El-Adly and Samen (2018). Additionally, the study introduces two new dimensions of perceived value: location value and hotel neighborhood value, both recognized by customer. Furthermore, 58 % of participants praised the quality of food at restaurants outside of the hotel, indicating a form of self-gratification

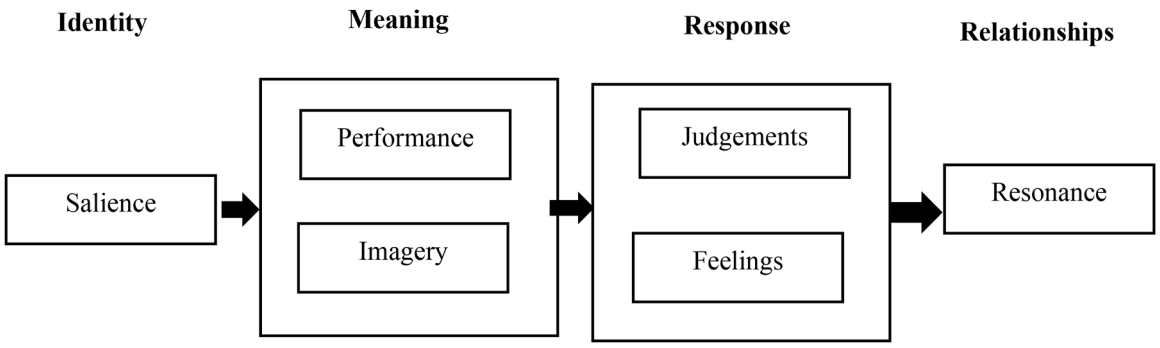


Fig. 1. Keller brand equity model.

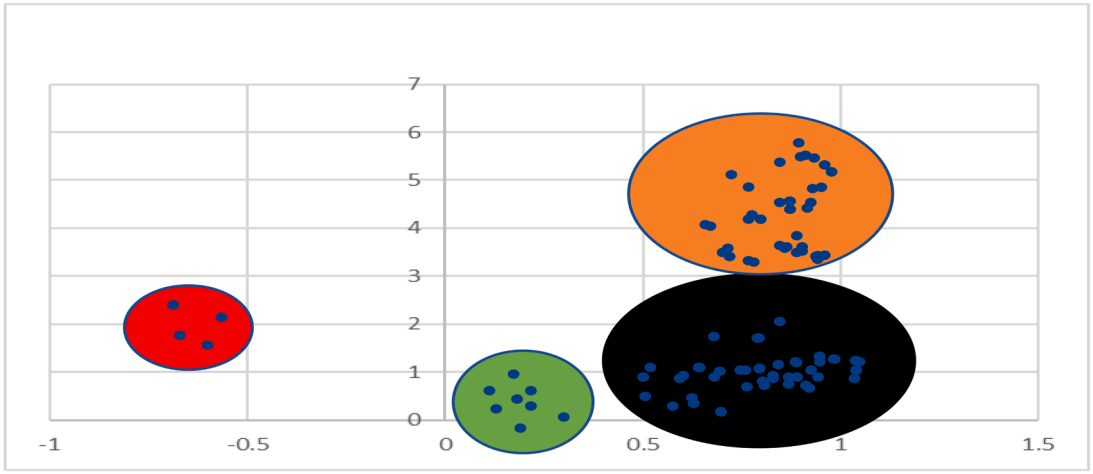


Fig. 2. Customer reviews clustering from document analysis.

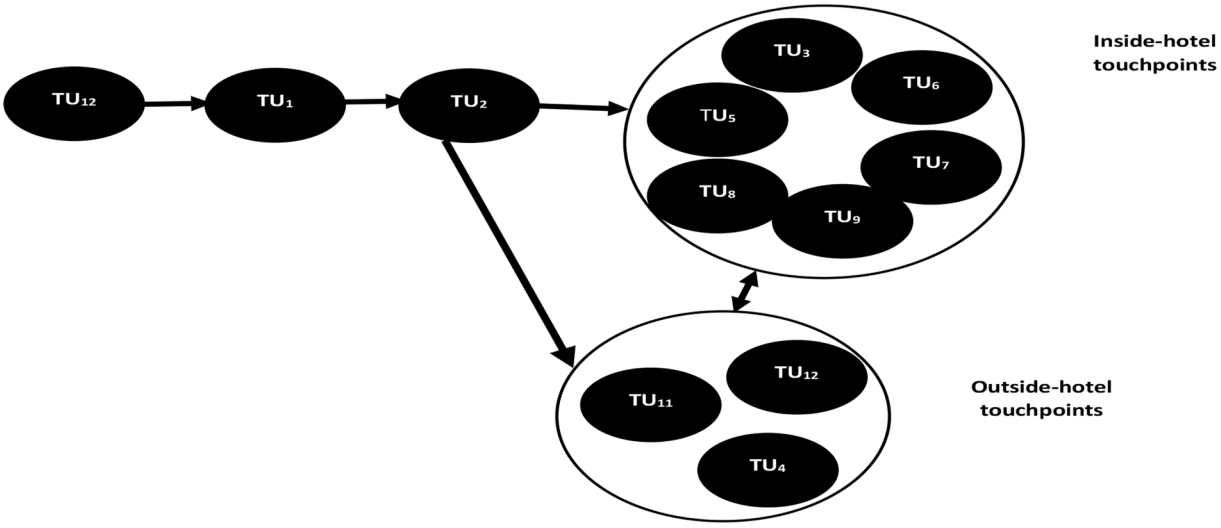


Fig. 3. Customer journey map via reviews.

value within the hotel restaurant (Davis & Hodges, 2012; El-Adly & Samen, 2018). In addition to shedding light on this relevance features also lend support to self-esteem theory and Maslow's hierarchy of needs. Self-esteem theory is essential for comprehending why customers share their knowledge and experiences with a brand, as positive feedback can boost their self-esteem and reinforce their positive self-concept (Orth & Robins, 2014). This emphasizes the significance of providing customers with a positive experience that makes them feel valued and appreciated. The findings also confirm the connection between the brand equity model and customer reviews, aligning with prior research by Kimpakorn and Tocquer (2010) and Liu et al. (2017).

From the perspective of the brand equity model analyzed with customer reviews, differences in the influence curves among hierarchical theoretical elements were observed. The data from reviews revealed a multi-path graph, indicating that in customers' perceptions, noteworthy aspects were mentioned directly. For instance, starting from the

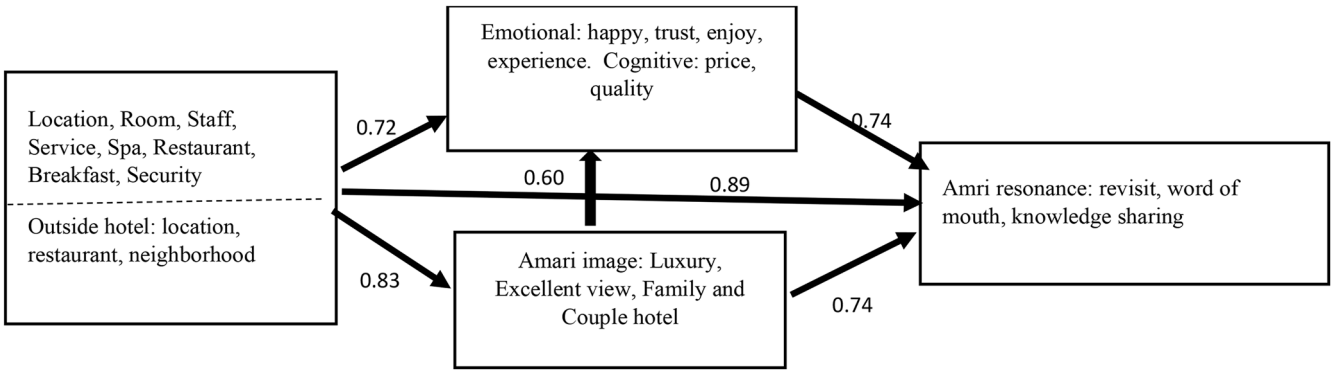


Fig. 4. Illustrates the synthesized Amari equity map based on the sample data with influence weight (probability).

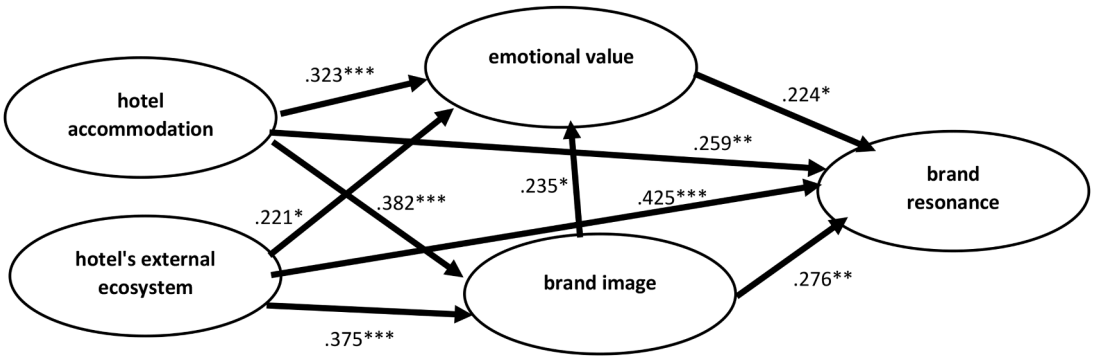


Fig. 5. The strength of causal paths using SEM.

"hotel accommodation" as a salient stage, there were influence links to "emotion," "brand image," and "brand resonance." This observation aligns with the appraisal theory (Fischer et al., 1990).

6.2. Managerial implications

This research offers several significant implications for hotel managers. Firstly, it presents a new and proactive strategy for acquiring knowledge of the buying process and past experiences from brand customers. By analyzing the customer journey map derived from narrative reviews, managers can gain insights into the mood and tone of consumers towards the brand. This can help identify the strengths and weaknesses of product and service attributes, ultimately leading to improvements in brand equity, an asset for any firm. Additionally, customer reviews often include comparisons with competitors, and by analyzing these comparisons, managers can gain a better understanding of the hotel's perception in the marketplace and identify areas for improvement. Moreover, this research introduces a new multidimensional construct of perceived value that encompasses the eco system value. By integrating these factors, managers can allocate resources more efficiently and gain a better understanding of the surrounding hotel environment. This provides valuable insights into different aspects of hotel brand equity, helping them make data-driven decisions to enhance the brand's competitiveness and reputation.

Apart from these benefits, this research can also aid in improving customer retention rates, a crucial advantage for hotel managers. By utilizing emotion and magnitude values, marketers can perform automated churn analysis to identify pain points and resolve them. This can lead to increased customer retention rates, a critical responsibility of marketers, ultimately strengthening the brand's reputation and competitiveness in the long run.

Overall, this research is conducted within a white-box framework, prioritizing cost efficiency for implementation within information systems. This approach aims to retain customers and enable informed decision-making for enhancing the brand's overall performance. The introduction of the multidimensional construct of perceived value enables managers to allocate resources more effectively and gain a comprehensive understanding of the hotel environment. Consequently, this can result in enhanced brand equity and a stronger competitive position in the marketplace.

6.3. Limitation and future research directions

The present study has several limitations that should be considered when interpreting the results. Firstly, the analysis of emotion and magnitude values is subject to cultural influence, as the data used in this research was collected in English, which may not be representative of other languages or cultures. Furthermore, some reviews only repeat positive experiences while failing to provide feedback on all touchpoints or services that were not used. This may limit the ability to construct a comprehensive customer journey map, as the summation of experiences may not accurately reflect the sequence of events. Finally, the use of natural language processing techniques may result in errors and the use of wildcards, which can affect the accuracy of the findings. Moreover, the evidence in the study is based on Amari, a luxury hotel in Phuket, which may differ from other accommodations.

To mitigate the limitations of this study, future research endeavors should encompass a broader spectrum of languages and cultures to probe potential variations in emotion and magnitude values. Furthermore, researchers should delve into strategies for incentivizing customers to offer feedback across all touchpoints, including those less frequently utilized, to craft a more comprehensive customer journey map. Enhancements in opinion mining, particularly through the incorporation of wildcards like "emoji" facilitated by electronic devices, should also be explored to refine text mining techniques.

6.4. Conclusion

In conclusion, this research has made significant theoretical and managerial contributions to the field. From a theoretical perspective, the examination of 250 customer reviews of Amari Hotels identified four distinct customer segments, with an impressive 90 % displaying high loyalty and a strong inclination to recommend the brand. These findings align with prior research on brand awareness and performance, emphasizing the hotel's positive reputation and brand equity. Additionally, the study highlights the crucial role of the hotel's location, introducing novel dimensions of perceived value and supporting self-esteem theory and Maslow's hierarchy of needs. The analysis also reveals differences in influence curves among hierarchical theoretical elements, aligning with the appraisal theory.

From a managerial standpoint, this research offers valuable insights for hotel managers. It presents a proactive strategy for acquiring customer knowledge and improving brand equity by analyzing customer journey maps derived from narrative reviews. Managers can identify strengths, weaknesses, and areas for improvement, as well as gain a better understanding of the hotel's perception in the marketplace through customer comparisons. The multidimensional construct of perceived value introduced in this study allows managers to allocate resources efficiently and enhance the brand's competitiveness and reputation.

Furthermore, this research can aid in improving customer retention rates by utilizing emotion and magnitude values for automated churn analysis, ultimately strengthening the brand's reputation and competitiveness. Overall, the cost-efficient white-box framework implemented in this study prioritizes customer retention and informed decision-making within information systems, contributing to enhanced brand performance and competitive positioning in the marketplace.

CRediT authorship contribution statement

Intaka Piriyakul: Writing – review & editing, Writing – original draft, Validation, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Shawanluck Kunathikornkit:** Writing – review & editing. **Rapepun Piriyakul:** Writing – original draft, Methodology, Conceptualization.

Declaration of competing interest

There were no potential conflicts of interest disclosed by the authors in relation to the research, authorship, and/or publication of this article.

Data availability

Supplemental material for this article, including the data source, is available online at https://www.tripadvisor.com.

Ethical Approval and Informed Consent

The research proposal received approval from the Research Center of the Faculty of Business Administration for Society, Srinakharinwirot University. The data used in the study was obtained from an open-source, publicly available source.

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